Pests & Disease

Integrated pest management for US crops — insects, diseases, weeds & more

A detailed reference guide • United States
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Note: *This guide compiles general, widely accepted agronomic practice for US growers. Conditions vary by state, soil, and season — confirm specifics with your local Cooperative Extension office (find yours at nifa.usda.gov/land-grant-colleges-and-universities-partner-website-directory) and always follow current product labels and local regulations.*

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1. Integrated Pest Management (IPM)

Integrated Pest Management is the framework that underlies all modern pest control. Rather than spraying on a fixed schedule, IPM combines prevention, monitoring, and a mix of control tactics to keep pests below economically damaging levels while minimizing cost, resistance, and harm to people, beneficial organisms, and the environment.

The IPM hierarchy

- **Prevention first.** Resistant varieties, crop rotation, sanitation, healthy soil, and proper planting dates stop many problems before they start.
- **Monitor and identify.** Scout regularly; correctly identify the pest and assess how many there are.
- **Decide using thresholds.** Act only when pest numbers justify the cost of control — not at the first sight of a pest.
- **Choose the least-disruptive effective tactic.** Cultural and biological controls first; chemical control when needed, chosen and timed to protect beneficials.
- **Evaluate.** Record what worked and refine next season.

2. Scouting & Monitoring

You cannot manage what you do not measure. Regular scouting is the engine of IPM — it tells you what is present, where, and in what numbers, so decisions are based on fact rather than fear.

- Walk fields at least weekly during active growth, more often at vulnerable stages or during pest outbreaks.
- Follow a representative pattern (W-shaped or zig-zag), checking multiple spots, not just field edges.
- Inspect whole plants — leaf undersides, growing points, stems, and roots — where pests hide.
- Use tools: sweep nets for foliage insects, sticky traps and pheromone traps for flying pests, and a hand lens for small ones.
- Keep dated records with location, pest, life stage, and counts to track trends and time controls.

3. Economic Thresholds

An economic threshold is the pest level at which the cost of damage begins to exceed the cost of control — the signal to act. Treating below threshold wastes money and can breed resistance and flare secondary pests by killing their natural enemies; waiting too far above it sacrifices yield.

Note: *Thresholds are crop-, pest-, growth-stage-, and price-specific. State Extension services publish current thresholds (e.g., number of insects per plant or per sweep, or percent defoliation). Always use locally validated numbers.*

A related idea is percent defoliation: many crops tolerate surprising leaf loss with little yield effect early in the season, but far less during flowering and fruit/grain fill. Knowing the tolerant and sensitive stages

prevents needless sprays.

4. Insect Pests of Major Crops

Field crops

Pest	Crops affected	Damage & signs
Corn rootworm	Corn	Larvae feed on roots (lodging); adults clip silks. Rotation and Bt hybrids manage it.
European corn borer	Corn	Boring in stalks/ears; broken tassels. Largely controlled by Bt corn.
Corn earworm / bollworm	Corn, cotton, soy, sorghum	Larvae feed in ears, bolls, pods.
Soybean aphid	Soybean	Colonies on new growth; sticky honeydew, stunting, virus spread.
Bean leaf beetle	Soybean	Round holes in leaves; pod feeding late season.
Stink bugs	Soy, cotton, many	Piercing damage to pods/bolls/fruit; deformed seed.
Aphids (various)	Most crops	Sap-sucking; honeydew, sooty mold, virus vectors.
Cutworms	Corn, veg seedlings	Cut young plants off at soil line at night.

Vegetable & specialty crops

Pest	Crops affected	Damage & signs
Colorado potato beetle	Potato, eggplant, tomato	Striped adults, red larvae; rapid defoliation.
Cabbage worms / loopers	Brassicas	Chewed leaves, holes; green larvae and frass.
Squash bug / vine borer	Squash, pumpkin, melon	Wilting vines; borers inside stems.
Cucumber beetle	Cucurbits	Feeding plus spread of bacterial wilt.
Tomato hornworm	Tomato, pepper	Large green caterpillar; heavy defoliation, fruit damage.
Flea beetles	Brassicas, eggplant, many	Tiny "shot-hole" pits in leaves, worst on seedlings.
Thrips	Onion, many	Silvery streaks, distorted growth; virus vectors.
Spider mites	Many (hot, dry)	Stippling, fine webbing, bronzing of leaves.

5. Common Plant Diseases

Diseases are grouped by the type of pathogen. Correct identification matters because fungal, bacterial, and viral diseases respond to very different controls — fungicides do nothing against bacteria or viruses.

Type	Examples	Typical signs	Management
Fungal	Gray leaf spot, rusts, powdery & downy mildew, white mold, early/late blight	Spots, pustules, mold, wilts	Resistant varieties, rotation, residue management, fungicides, airflow
Bacterial	Bacterial blight, bacterial spot, bacterial wilt	Water-soaked lesions, ooze, sudden wilt	Clean seed, sanitation, copper products, resistant varieties; no cure once systemic
Viral	Mosaic viruses, yellows, leaf curl	Mottling, distortion, stunting	Control insect vectors (aphids, thrips), remove infected plants, resistant varieties
Nematode / other	Root-knot, cyst nematodes	Galls or cysts on roots, stunting, patchy fields	Rotation, resistant varieties, soil testing

Note: *The disease triangle — a susceptible host, a pathogen, and favorable environment — must all be present for disease. Most cultural controls work by removing one leg: resistant hosts, sanitation to reduce pathogen, and airflow/dry foliage to make the environment unfavorable.*

6. Weeds as Pests

Weeds are often the costliest pest in field crops, competing for light, water, and nutrients and hosting insects and diseases. Herbicide-resistant weeds (notably waterhemp, Palmer amaranth, marestail, and ryegrass) make diversified management essential.

- Identify key weeds and know whether they are annual, biennial, or perennial — this dictates control timing.
- Control weeds early: the critical weed-free period right after crop emergence protects most of the yield.
- Rotate herbicide modes of action and combine with tillage, cover crops, narrow rows, and clean equipment to slow resistance.
- Never let escapes go to seed — one Palmer amaranth plant can produce hundreds of thousands of seeds.

7. Nematodes

Plant-parasitic nematodes are microscopic roundworms that attack roots, causing stunting, yellowing, wilting, and patchy, declining fields that are easily mistaken for nutrient or water problems. Soybean cyst nematode and root-knot nematode are among the most economically important in the US.

- Confirm with a soil and root assay — symptoms alone are not diagnostic.
- Rotate to non-host crops and plant resistant varieties where available.
- Maintain soil organic matter and avoid moving infested soil on equipment.

8. Vertebrate Pests

Deer, rodents (voles, gophers, mice), rabbits, birds, and groundhogs can cause significant local damage, especially to young plantings and at field edges.

- **Exclusion** is the most reliable tool: fencing for deer and rabbits, netting for birds, hardware cloth around trunks for rodents.
- **Habitat management** — mowing field margins and removing brush — reduces rodent cover.
- **Scare and repellent** tactics give short-term relief but animals habituate; rotate methods.
- Follow all wildlife regulations; many species and control methods are legally restricted.

9. Control Methods

Method	What it is	Examples
Cultural	Growing practices that make conditions unfavorable to pests	Rotation, sanitation, planting date, resistant varieties, spacing, water management
Mechanical / physical	Physically removing or blocking pests	Cultivation, hand-removal, traps, row covers, mulches, barriers
Biological	Using natural enemies to suppress pests	Lady beetles, lacewings, parasitic wasps, predatory mites, Bt, beneficial nematodes
Chemical	Pesticides, used selectively within IPM	Insecticides, fungicides, herbicides — chosen, timed, and rotated to minimize harm and resistance

No single method is sufficient alone. IPM layers them, leaning on prevention and biology and reserving chemicals for when monitoring shows they are warranted.

10. Beneficial Organisms

Many insects and organisms are allies that quietly suppress pests. Protecting them — by avoiding broad-spectrum sprays and providing habitat and flowering insectary plants — is a cornerstone of sustainable pest control.

Beneficial	Controls
Lady beetles (ladybugs)	Aphids, mites, soft-bodied insects, eggs
Lacewing larvae	Aphids, thrips, mites, small caterpillars
Parasitic wasps (e.g., Trichogramma)	Caterpillar and aphid hosts (laid inside them)
Ground & rove beetles	Soil-dwelling larvae, slugs, weed seeds
Predatory mites	Spider mites, thrips
Pollinators (bees)	Not pest control, but vital — protect from spraying

Note: *To protect pollinators and natural enemies, avoid spraying during bloom, spray in the evening when bees are not foraging, and choose selective products over broad-spectrum insecticides whenever possible.*

11. Pesticide Selection & Safety

When a pesticide is warranted, selecting and applying it correctly protects efficacy, your safety, and the environment, and keeps you legal.

- **Identify first.** Match the product to the correctly identified pest and crop; the wrong product wastes money and may harm the crop.
- **Read and follow the label.** The label is the law — it dictates rate, timing, protective equipment, re-entry interval, and pre-harvest interval.
- **Rotate modes of action.** Using the same chemistry repeatedly selects for resistance; rotate IRAC/FRAC/HRAC groups.
- **Time applications** to the pest's vulnerable stage and to protect pollinators and beneficials.
- **Wear required PPE** and observe re-entry (REI) and pre-harvest (PHI) intervals.
- **Protect water.** Respect buffer zones, avoid spraying before heavy rain, and prevent drift and runoff.
- **Store and dispose** of products and containers safely and legally; keep application records.

Note: *Pesticide registrations, rates, and restrictions change. Always verify against the current product label and your state regulations before use, and consult your Cooperative Extension office for locally recommended products and thresholds.*